

MLDM: Multiple-Location Disease Mapping

MLDM_Example

[1] Simulate data from the proposed model:

- Setting the reproducibility seed and initializing packages for data simulation:

```
set.seed(2347)

library(MLDM)
library(mnormt)           #Multivariate normal distribution
library(clusterGeneration) #Random correlation matrix

## Loading required package: MASS

• Setting the global data values:
c<-3 #Number of disease groups

n<-rpois(n = c,
        lambda = 50) +
  1 #Number of individuals per group

m<-rpois(n = sum(n),
        lambda = 10) +
  1 #Number of locations per individual

n_grid<-7
d<-n_grid^2 #Total number of regions

#Queen's adjacency matrix
R<-matrix(0,
          nrow = d,
          ncol = d)
index<-function(i, j){
  return((i - 1)*n_grid + j)
}
for(j in 1:n_grid){
  for(k in 1:n_grid){
    idx<-index(j, k)
    neighbors<-c()
    if(j > 1) neighbors<-c(neighbors, index((j-1), k))
    if(j < n_grid) neighbors<-c(neighbors, index((j+1), k))
    if(k > 1) neighbors<-c(neighbors, index(j, (k-1)))
    if(k < n_grid) neighbors<-c(neighbors, index(j, (k+1)))
    if((j > 1) && (k > 1)) neighbors<-c(neighbors, index((j-1), (k-1)))
    if((j > 1) && (k < n_grid)) neighbors<-c(neighbors, index((j-1), (k+1)))
    if((j < n_grid) && (k > 1)) neighbors<-c(neighbors, index((j+1), (k-1)))
    if((j < n_grid) && (k < n_grid)) neighbors<-c(neighbors, index((j+1), (k+1)))
    R[idx, neighbors]<-1
  }
}
R_full<-R
R<-R_full[-1, -1] #Reference region is [1,1]

p_v<-2 #Number of region-level covariates
```

```

v_full<-matrix(rnorm(n = (d*p_v)),
               nrow = d,
               ncol = p_v)
v<-v_full[-1,] #Remove reference region row

p_x<-1 #Number of individual-level covariates
x<-as.matrix(scale(rnorm(n = sum(n)))) #Standardized individual-level covariate

```

- Setting the values for the statistical model parameters:

```

rho_true<-0.80
Q_true<-rho_true*(diag(rowSums(R)) - R) +
  (1.00 - rho_true)*diag(d - 1)
Q_inv_true<-chol2inv(chol(Q_true))

Sigma_true<-rcorrmatrix(c*(1 + p_x)) #True cross-covariance (beta and delta jointly)

gamma_true<-rnorm(n = (c*(1 + p_x)*p_v)) #N(0,1) region-level covariate coefficients

v_design_list<-vector("list", (d - 1))
for(j in 1:(d-1)){
  v_design_list[[j]]<-as.matrix(Matrix::bdiag(replicate(c*(1 + p_x),
                                                         matrix(v[j,], 1),
                                                         simplify = FALSE)))
}
v_design<-do.call(rbind, v_design_list)

#Draw theta_true (beta and delta jointly) from MCAR prior
theta_true<-rmnorm(n = 1,
                  mean = (v_design%*%gamma_true),
                  varcov = (Q_inv_true%x%Sigma_true))
theta_true<-matrix(theta_true,
                  nrow = c*(1 + p_x),
                  ncol = (d - 1))

#Sum-to-zero centering (matches on-the-fly centering in the sampler)
mean_struct<-matrix((v_design%*%gamma_true),
                    nrow = c*(1 + p_x),
                    ncol = (d - 1))
theta_true<-theta_true -
  matrix(rowMeans(theta_true - mean_struct),
        nrow = c*(1 + p_x),
        ncol = (d - 1),
        byrow = FALSE)

beta_true<-theta_true[1:c,] #c x (d-1)
delta_true<-theta_true[(c + 1):(c*(1 + p_x)),]  #(c*p_x) x (d-1)

sigma2_true<-0.10^2 #Individual-level random effect variance (shared across groups)

```

- Simulating the analysis dataset:

```

#Group indicator matrix (used for simulation only; built internally by MLDM)
X<-matrix(0,
          nrow = sum(n),

```

```

        ncol = c)
counter<-0
for(j in 1:c){
  X[c((1 + counter):(n[j] + counter)), j]<-1
  counter<-counter + n[j]
}

#Blocked individual covariate matrix (used for simulation only; built internally by MLDM)
G<-matrix(0,
          nrow = sum(n),
          ncol = (c*p_x))
counter<-0
for(j in 1:c){
  G[c((1 + counter):(n[j] + counter)),
    c(((j-1)*p_x + 1):(j*p_x))]<-x[c((1 + counter):(n[j] + counter)),]
  counter<-counter + n[j]
}
X_star<-cbind(X, G)

alpha_true<-matrix(rnorm(n = (sum(n)*(d - 1)),
                        sd = sqrt(sigma2_true)),
                  nrow = sum(n),
                  ncol = (d - 1))

denom<-exp(X_star%%theta_true[,1] + alpha_true[,1])
for(j in 2:(d-1)){
  denom<-denom + exp(X_star%%theta_true[,j] + alpha_true[,j])
}
p<-matrix(NA, nrow = sum(n), ncol = d)
p[,1]<-1.00/(1.00 + denom)
for(j in 1:(d-1)){
  p[, (j+1)]<-exp(X_star%%theta_true[,j] + alpha_true[,j])/(1.00 + denom)
}

z<-matrix(NA, nrow = sum(n), ncol = d)
for(j in 1:sum(n)){
  z[j,]<-rmultinom(n = 1,
                  size = m[j],
                  prob = p[j,])
}

#True residual spatial signal
denom<-exp(beta_true[,1])
for(j in 2:(d-1)){
  denom<-denom + exp(beta_true[,j])
}
p_signal_true<-matrix(NA, nrow = c, ncol = d)
p_signal_true[,1]<-1.00/(1.00 + denom)
for(j in 1:(d-1)){
  p_signal_true[, (j+1)]<-exp(beta_true[,j])/(1.00 + denom)
}

```

[2] Fit MLDM to identify disease group-specific spatial hotspots:

```

results<-MLDM(mcmc_samples = 11000,
              z = z,
              n = n,
              m = m,
              v = v,
              R = R,
              x = x,
              metrop_sd_rho_trans = 0.75)

```

```

## Progress: 10%
## rho Acceptance: 38%
## *****
## Progress: 20%
## rho Acceptance: 37%
## *****
## Progress: 30%
## rho Acceptance: 36%
## *****
## Progress: 40%
## rho Acceptance: 36%
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## Progress: 50%
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## Progress: 60%
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## Progress: 70%
## rho Acceptance: 36%
## *****
## Progress: 80%
## rho Acceptance: 36%
## *****
## Progress: 90%
## rho Acceptance: 36%
## *****
## Progress: 100%
## rho Acceptance: 36%
## *****

```

[3] Analyzing Output:

```

burnin<-1000
keep_set<-seq((burnin + 1), 11000, 1)

#p_signal: posterior means vs. true values (all groups and regions)
p_signal<-p_signal_space_create(results$beta[, , keep_set])
p_signal_pm<-apply(p_signal, c(1, 2), mean) #c x d posterior mean matrix

plot(as.vector(p_signal_true),
     as.vector(p_signal_pm),
     pch = 16,
     xlab = "True p_signal",
     ylab = "Posterior Mean p_signal")
abline(0, 1,

```

```
col = "red",  
lwd = 2)  #45-degree reference line
```

